Math and Theory: collection overview

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Research process and AI disclosure

This collection reports fully autonomous agentic research. Dev D. Goyal is responsible for publication. Models used: GPT 6 Astra by OpenAI in ChatGPT and Codex, GPT 6.1 Sol by OpenAI in Codex, and Opus 5.5 by Anthropic used with Claude Code. Each paper records how they were used.

Opus 5.5 by Anthropic used with Claude Code assessed the October 2026 uploads on ADMM multiplier steps, Nesterov flows, Chambolle–Pock cycles, critical hard-core correlations, Schmidt numbers of the Tiles state, a binary spectral tile, the Keller–Klein conjecture, unit square classes of totally real fields, depth-one Artin–Schreier towers, the upper PPT entanglement-cost hierarchy, matrix-pencil rank distance, selfadhesivity of polymatroids, nonnegative rank of Kronecker squares, multi-marginal maximal monotonicity, the Berge-triangle saturation spectrum, three-uniform Berge-path saturation, three-uniform Berge-C_4 saturation, Singer-invariant blocking sets of $\mathrm{PG}(2, 13)$ and order polynomials of width-two posets and the November 2026 uploads on Berge saturation of the subdivided star on sharp stability and exact cap optima for Rademacher sums and on finite real counterexamples to q-permanent monotonicity, and on graphs with five harmonic boundary vertices, on CHSH equality, rigidity and full-ellipsoid envelopes at fixed steering-ellipsoid centre, on optimal entanglement witnesses under ellipsoid containment, on the full correlation range, filtering frontier and volume trade-off at fixed steering centre, on one-sided filtering of the two-qubit F_3 functional, on the one-sided CHSH measure under local filtering and on saturating the steering-centre bound on the fully entangled fraction, coordinated independent AI referee checks, extended the planar Nesterov result, classified and formalized the complements of the binary tile, formalized the Berge-path counterexample family and the Berge-C_4 upper construction, ported the Lean proof of the subdivided-star saturation formula, wrote the Lean proof of the blocking-set result, checked the width-two certificates with its own program and in Lean for chains of size at most four, determined the q-permanent certificate independently by modular arithmetic and removed native evaluation from its Lean development, rechecked the harmonic-boundary catalogues with independent enumerations and assembled their Lean files into one audited project, separated the CHSH steering-centre results from the earlier public record and checked them with its own exact and numerical programs, added whole-project axiom audits to the ellipsoid-witness and one-sided CHSH Lean developments, compared the one-sided F_3 formulas with a brute-force filter search and rebuilt their Lean project with a whole-project audit, separated the fully-entangled-fraction results from the earlier public record of the bound and rebuilt that Lean project from source with a whole-project audit, ran fresh Lean replays, wrote or revised their twenty-nine manuscripts, and updated this overview and the research website. The research in those uploads was produced by autonomous AI agents; apart from the hard-core, tiling and Keller–Klein packages, which record ChatGPT sessions, the unit-squareclass package, whose tool logs indicate OpenAI Codex, the multi-marginal Minty package, which records an OpenAI GPT-6 Astra workflow, the Berge-path package, which attributes its work to an Astra agent workflow, the Berge-C_4 package, whose research contract specifies Astra referees for its internal reviews, the blocking-set package, which

describes its agents as Astra agents, the width-two package, whose records name a GPT-6 Astra research workflow, the subdivided-star package, whose records describe its internal review as an Astra review, and the Rademacher stability package, whose records describe an autonomous GPT-6 Astra Pro session, their records do not name the system used.

AI checks are not expert human validation. These preprints have not been externally peer reviewed, and the collection makes no claim of first priority. Correspondence received after publication brought earlier or overlapping work on six results to light, and a check against public problem records on 31 October 2026 found earlier or related public work for eight more; the priority and novelty record documents when it was learned and what remains uncertain. The authorship disclosure identifies the tools and their roles.

Abstract

The catalogue contains sixty manuscripts and research notes across 59 result families in mathematics, theoretical computer science, and mathematical physics. Each paper states its formal verification scope. The thermal proof covers finite spectral statements, without a formal stochastic long-time identification theorem. The Gyires preprint proves inequalities for structured matrices and develops related polynomial bounds. Its principal order-five and order-six claims, the all-order permutation-polynomial and four-block theorems, and the exceptional-vector theorem have complete Lean proofs. The results comprise counterexamples, exact special cases, sharp sequence bounds, and convergence thresholds for optimization methods. The collection makes no claim of first priority. The papers have not been peer reviewed.

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1 How to use the collection

The manuscript catalogue links each paper to its formal statements and independent computations. The verification guide records the commands and their results. Seven families share a Lake project pinned to Lean 4.35.0. Its 29 selected Comparator statements include equality, sharpness, and auxiliary positivity claims. The decision-tree and thermal packages use separate Lean 4.19.0 environments and a pinned Mathlib revision. The decision-tree package has 29 axiom reports of its own. This is a separate count from the shared Comparator selection.

2 Mathematics

2.1 Matrix theory

2.1.1 Real positive-definite counterexamples to extended q-permanent monotonicity

For every real parameter $q > 1$, some real symmetric positive-definite matrix has a negative q-permanent derivative at q. An explicit matrix of order five decreases from $q = 14$ to $q = 15$. The construction also gives correlation matrices with no zero entries and integer-entry examples. The dimension can depend on q. The real-symmetric question on $[-1, 1]$ remains outside the result.

Scope: Counterexample and all-parameter extension.

Manuscript Formal statement and prior work.

2.1.2 Finite real counterexamples and spectral obstructions for q-permanent monotonicity

An explicit integer Gram matrix B of order 896 and rank three, with row coordinates at most 32 in absolute value, has a negative derivative of its inversion-weighted q-permanent at $q = 1$, by the exact integer comparison $1000U > 1001\binom{896}{2}\operatorname{per}(B)$. The integer matrix $A = MB + I$ is positive definite, its q-permanent decreases between explicit rationals $0 < q_1 < q_2 < 1$, and identity padding gives such counterexamples in every order $N \geq 896$. This answers the real-symmetric case of the Bapat–Lal monotonicity conjecture on $[-1, 1]$ negatively and refutes da Fonseca's conjectures inside the original interval. Fixed-rank spectral and ordering obstructions are also proved. Lean 4.19.0 proves the endpoint identity, certificate soundness for every type list, the reductions to rational positive-definite counterexamples in every order $N \geq 896$ and the rank of the frozen input, with no native evaluation; the integer comparison itself is checked by two independent exact programs, and the explicit regularization and the asymptotic theorems are conventional.

Manuscript Proof and reproduction.

2.1.3 Gyires's permanent inequality for structured matrices

Every nonnegative real circulant matrix A of order five satisfies $24\operatorname{per}(AA^T) \leq 625\operatorname{per}(A)^2$. Equality holds exactly for constant matrices. This proves the order-five circulant case of Gyires's ratio inequality. A bound in sorted-gap coordinates has sharp coefficient 484. The conjecture for arbitrary doubly stochastic matrices remains open.

Scope: Structured matrix inequalities and related polynomial bounds. The principal order-five and order-six claims, the all-order permutation-polynomial and four-block theorems, and the exceptional-vector theorem have Lean proofs.

Manuscript Formal statement and prior work.

Broader arguments in the same preprint

The manuscript also proves results for circulants of orders six and seven, with sharp sorted-gap margins. It treats $aI + bQ + cQ^2$ in every dimension, where Q is any permutation matrix and the coefficients are nonnegative. Further results cover four-block doubly stochastic matrices with unequal block sizes and boundary entries, matrices with one exceptional column or row, and every rank-at-most-two doubly stochastic matrix through order fifteen.

The supporting sections prove structural, scalar, deletion, balancing, local polynomial, and configuration-model results. Exact counterexamples identify the limits of auxiliary assertions. The complete support record gives the source map and exact reproduction commands.

The all-order four-block and exceptional-vector theorems have complete Lean proofs. Their inequalities and equality cases are among the 29 selected formal statements. The rank-two theorem has a complete Lean proof for orders 2 through 8. The general order-seven circulant theorem and rank-two orders 9 through 15 remain incomplete in Lean. Separate Lean packages prove the principal order-six and all-order permutation-polynomial theorems. The all-order rank-two problem and the unrestricted Gyires conjecture remain unproved here.

2.1.4 A polygon enclosure fails for sums of normal matrices

A polygon enclosure fails for a specified pair of normal matrices and its complete transposition graph. Actual normal matrices, determinants, straight-segment exterior and component exclusion are formalized. Correspondence with an external source conjecture is conditional.

Manuscript Reproduction and source context.

2.1.5 An unbounded obstruction to projection AM–GM in Loewner order

Three rank-one orthogonal projections in dimension three violate the symmetrized-product Loewner AM–GM bound. No finite real multiplier repairs it, even with a positive-definite mean. The examples do not satisfy the additional nonsingular block structure of the motivating coordinate-descent application.

Formal scope: Lean verifies a rational three-projection matrix obstruction and a parameterized obstruction for every real multiplier in dimension three; rank-one, positive-definite mean, rational all-multiplier choices, and the standard minimal-factor observation have complete conventional proofs.

Manuscript Proof and reproduction.

2.1.6 Matrix-pencil rank distance as an explicit completion problem

Over every infinite field, two matrix pencils have strict-equivalence classes at normal-rank distance at most x exactly when an explicit dilation of the first is a constant full-rank compression of the dilation of the second plus x scalar blocks. With a cited block minimal bases theorem this proves Conjecture 1 of Dodig and Stošić: the bounded rank perturbation problem for matrix pencils is a completion problem. The explicit theorem is formalized in Lean 4.35.0-rc3.

Manuscript Proof and reproduction.

2.1.7 Counterexamples to the Kronecker-square lower bound for nonnegative rank

For every $k \geq 0$ an explicit rational nonnegative matrix C of order $k + 11$, rank $k + 10$ and nonnegative rank $k + 11$ satisfies $\operatorname{rank}_+(C \otimes C) \leq (k + 10)^2 + 9$. This disproves the conjecture of Vandaele, Gillis, Glineur and Tuyttens that $\operatorname{rank}_+(M \otimes M) \geq \operatorname{rank}_+(M) \operatorname{rank}(M)$. The family is formalized in Lean 4.35.0-rc3.

Manuscript Proof and reproduction.

2.2 Graph theory and graph polynomials

2.2.1 An entropy increase for a structurally disassortative DomiRank graph

A connected graph with 14 vertices and 32 edges has negative degree–log-degree edge covariance. All resolvent scores are positive on $[251/1000, 63/250]$ inside the stable domain. Its entropy nevertheless increases by more than $1/10000$ in natural-logarithm units. This refutes the source's assertion for every positive-score interval.

Scope: Counterexample to an interval monotonicity conjecture.

Manuscript Formal statement and prior work.

2.2.2 Domination sequences of powers of paths and cycles

The domination counts of every natural power of a path or cycle form an ultra-log-concave sequence. Dividing each count by its binomial coefficient gives the normalized log-concavity inequality. The proof includes all natural graph orders and powers. It does not assert the same property for arbitrary graphs or real-rootedness of their domination polynomials.
Manuscript Formal proof and reproduction.

2.2.3 Sharp escape of complex domination roots in square flowers

Sharp complex-root escape for square flowers, with complete flower and book real-root classifications and multiplicities. All numbered manuscript conclusions are formalized in seventy modules with 258 endpoint audits. Historical novelty and exact external-source correspondence remain provisional.

Manuscript Reproduction and source context.

2.2.4 Leaf deletion can move the unique mode of a tree domination polynomial by two

Deleting a leaf can move the unique mode of an explicit tree domination polynomial by two. The fixed tree, deletion, actual domination counts and unique-mode movement are formalized. External question identity and historical priority remain provisional.

Manuscript Reproduction and source context.

2.2.5 The upper saturation spectrum of Berge triangles in triple systems

For every $n \geq 8$, a Berge-triangle-free triple system on n vertices with more than $\lfloor (n-1)^2/8 \rfloor$ edges is a spanning subsystem of a paired product $F(n,p)$. Saturated systems above this threshold are therefore paired products, and the Berge-triangle saturation spectrum there consists exactly of the sizes $p(n-2p)$ up to $\lfloor n^2/8 \rfloor$, with $n/4 - O(\sqrt{n})$ missing sizes. The threshold is sharp: a saturated system with exactly $\lfloor (n-1)^2/8 \rfloor$ edges that is not a paired product exists for every $n \geq 8$. Lean 4.35.0 proves that paired products are Berge-triangle-free and saturated, proves the construction for every $n \geq 8$ and the arithmetic of the window, and derives the spectrum statements from the containment theorem, which has a conventional proof and exact checks.
Manuscript Proof and reproduction.

2.2.6 Exact three-uniform Berge-path saturation numbers and a counterexample to a conjectured formula

Two disjoint loose 8-cycles with two isolated vertices form a Berge-P_{10}-saturated three-uniform hypergraph on 34 vertices with 16 hyperedges, one below the value conjectured by English,

Graber, Kirkpatrick, Methuku and Sullivan. For all $m \geq 10$ and $n \geq b = 2a_m + 1$ the saturation number equals the conjectured value except at the orders $b < n < 2b$ with $n \equiv 2 \pmod{2(m-2)}$, where it is one less and the extremal hypergraph is unique. Lean 4.35.0 verifies the counterexample family and the failure of the conjecture for every $m \geq 10$; the lower bound is conventional.
Manuscript Proof and reproduction.

2.2.7 Three-uniform Berge-C_4 saturation: linear bounds with coefficients $5/8$ and $3/4$, the cone spectrum and exact small values

For the three-uniform Berge-C_4 saturation number $s(n)$, in which any absent triple may be added, $\max(\lceil (n-1)/2 \rceil, \lceil (5n-3623)/8 \rceil, \lceil (5n+89)/8 - \sqrt{270n+1891}/4 \rceil) \leq s(n) \leq \lceil (3n-9)/4 \rceil$ for every $n \geq 9$, so the linear coefficient lies between $5/8$ and $3/4$; the previous explicit upper bound was $n-2$. Saturated cones are classified, with sizes exactly the integers from $\lceil (3n-5)/4 \rceil$ to $n-2$ and $n-1$ when $n \equiv 1 \pmod 3$, and exhaustive certificates give $s(n)$ for $n \leq 13$. Lean 4.35.0 verifies the upper construction for every $n \geq 9$ and the values for $n \leq 5$; the lower bound and the cone results are conventional.
Manuscript Proof and reproduction.

2.2.8 Exact Berge saturation numbers of the subdivided star in every uniformity and its minimality among trees

Let B_a be the star $K_{1,a}$ with one edge subdivided. For all $a, r \geq 3$ and $n \geq (r-1)a + 1$ the r-uniform Berge saturation number, in which any absent r-set may be added, is $\mathrm{sat}_r(n, \text{Berge-}B_a) = \lfloor a(n+1)/((r-1)a+1) \rfloor$, attained by disjoint loose stars, and saturated hypergraphs near the minimum are loose stars apart from a bounded number of vertices. For $a \geq 5$ every other tree T on $a+2$ vertices has $\mathrm{sat}_3(n, \text{Berge-}T) \geq \lceil (a+1)(n-2)/(2a+3) \rceil$, so B_a is the unique tree of its order with the least asymptotic lower three-uniform saturation coefficient, $a/(2a+1)$: the Berge analogue of a theorem of Faudree, Faudree, Gould and Jacobson. Lean 4.35.0 verifies the exact formula, with the saturation number defined as a minimum over all hypergraphs on n labelled vertices, and the near-minimum structure theorem; the smaller orders, the sharpness constructions and the minimality among trees are conventional.
Manuscript Proof and reproduction.

2.2.9 Graphs with five harmonic boundary vertices, subcubic graphs with six, and minimum-boundary two-trees

A vertex x of a connected graph is a Steinerberger boundary vertex if some vertex is strictly farther from x than its average distance from the neighbours of x. Every two-connected graph with at most five boundary vertices has diameter at most two, and a connected graph has exactly five boundary vertices if and only if it is a bridge subdivision of one of 227 reduced graphs of order at most fourteen, answering Problem 6.1 of Chiem, Dudarov, Lee, Lee and Liu. The proof reduces boundary metrics of every diameter, through the three cones of five-point metrics and a cap preserving local configurations, to an exact certificate leaving fifteen metrics, excluded by hand. Connected graphs of maximum degree three with exactly six boundary vertices are the bridge subdivisions of 201 reduced graphs. For $n \geq 6$ every two-tree on n vertices has at least $\lceil n/2 \rceil + 2$ boundary vertices, with three extremal two-trees at each even $n \geq 8$ and, at odd $n \geq 9$, a period-eight quartic quasi-polynomial number of them, asymptotic to $n^4/192$. Lean proves the degree-three obstruction at graph level, the 835 local infeasibility systems of the certificate and further components; the cone and capping reduction and the catalogue enumerations are conventional proofs and exact computations, rechecked by independent enumerations.
Manuscript Proof and reproduction.

2.3 Inequalities and real analysis

2.3.1 Diagonal concavity does not imply concavity for sums of elementary symmetric polynomials

For $M > 0$, let $f_M = 1 + M\sigma_1 + \sigma_2 + M\sigma_3 + \sigma_4$ in four variables, where σ_j is the elementary symmetric polynomial of degree j. The fourth root $f_M^{1/4}$ is concave on the positive diagonal exactly when $M \geq 1$. It is concave on the full positive orthant exactly when $M = 1$. Thus every $M > 1$ disproves Lachaume's criterion with strictly positive coefficients.

Scope: Counterexample and complete parameter classification.

Manuscript Formal statement and prior work.

2.4 Integer sequences and combinatorics

2.4.1 Alternating Motzkin sums

The principal root bound and bounded-forcing companion form one sequence family.

Sharp root log-concavity of alternating Motzkin sums Let $T_n = \sum_{j=0}^{n}(-1)^{n-j}M_j$, where M_j is the Motzkin number. The sequence $T_n^{1/n}$ is strictly log-concave on its tail starting at five. The inequality reverses at center five, so no earlier tail is log-concave. This strengthens Zhao's proposed tail starting at nine. The source recurrence and finite sum are linked in Lean.

Scope: Conjecture resolution with an optimal threshold.

Manuscript Formal statement and prior work.

Sharp log-concavity for alternating Motzkin sums A robust bounded-forcing recurrence theorem and corrected scaled corollary, with applications to alternating Motzkin-type sums. The arbitrary-real recurrence theorem and scaled corollary, including zero forcing, are formalized. Concrete sequences, uniform-family applications, root transfer and threshold classification retain conventional proofs.

Manuscript Reproduction and source context.

2.4.2 Counterexamples to a late-iterate ordering conjecture for generalized Hofstadter functions

For every $k \geq 4$, some input n satisfies $F_k^{\circ(2k+1)}(n) > F_{k+1}^{\circ(2k+2)}(n)$. Here the superscript denotes repeated composition. The proof constructs inverse iterates with exact gap $k-3$. The witness $F_4^{\circ 9}(132) = 8 > 7 = F_5^{\circ 10}(132)$ gives the first instance. The result concerns infinitely many parameters. It does not assert infinitely many violating inputs for each fixed parameter.

Scope: Counterexamples for every parameter k≥4.

Manuscript Formal statement and prior work.

2.4.3 A positive constant-recursive sequence with nonquasipolynomial power ranks

Exact minimal recurrences for every power of a rank-thirteen sequence, nonquasipolynomial ranks, and infinitely many power-rank profiles at each rank at least three. The main and arbitrary-rank companion theorems are formalized. Generating-function and polynomial corollaries retain conventional proofs.

Manuscript Reproduction and source context.

2.4.4 Order-polynomial positivity for width-two posets with a chain of at most five elements

If a finite poset P is the union of two chains A and B with $|A| \leq 5$, the chain B being arbitrarily long and the comparabilities between the chains arbitrary, then $|P|!\,\Omega_P(t)$ has nonnegative coefficients, a case of Conjecture 7.1 of Ferroni, Morales and Panova; moreover $[t]\,\Omega_P(t) \geq (a-1)!\,(b-1)!\,m/(a+b)!$, where m counts the elements of B comparable with some element of A. On each of 64 chambers the normalized linear coefficient is a polynomial of degree at most a in the gaps between boundaries, certified nonnegative by exact square certificates, and the ideal–filter reduction of Ferroni, Morales and Panova lifts it to all coefficients. Lean 4.35.0 checks the certificates for $|A| \leq 4$; the link with the order polynomial, the size-five chambers and the reduction are conventional or checked by an independent program. Public partial results on ordinal sums of one- and two-element antichains and on single cross-link factors cover different classes.
Manuscript Proof and reproduction.

2.5 Finite geometry

2.5.1 Blocking sets of PG(2,13) invariant under a fixed-point-free Singer subgroup meet some line in five points

Let H be the subgroup of order three of a Singer cycle group of $\mathrm{PG}(2, 13)$; it fixes no point. Every H-invariant blocking set meets some line in at least five points, and the 21-point blocking set of Blokhuis, Brouwer and Wilbrink attains five, so five is the least possible largest line intersection of an H-invariant blocking set; the same holds for every fixed-point-free collineation of order three. This settles a symmetric case of the plane of order 13 in Erdős Problem 1159. Lean proves the theorem for the specified subgroup, including an exhaustive search evaluated in the kernel with a soundness proof.
Manuscript Proof and reproduction.

2.6 Quadratic forms and number theory

2.6.1 A rank bound for represented totally positive unit squareclasses

A totally positive definite quadratic lattice of rank n over a totally real algebraic field, with integral norms but not necessarily classical, represents at most n classes of totally positive units modulo squares of units. This answers Question 3.4 of Daans, Kala, Man, Widmer and Yatsyna negatively; the bound is a short consequence of standard facts on finite reflection groups. The theorem is formalized in Lean 4.19.0 for all algebraic totally real fields.
Manuscript Formal proof and reproduction.

2.7 Valuation theory

2.7.1 Depth-one Artin–Schreier defect towers are composita

Over a perfect henselian subfield K of $k((y^{\mathbb{Q}}))$ containing $k(y)$ and without finite accumulation points, a nontrivial finite Artin–Schreier tower of depth one is a compositum of Artin–Schreier extensions of K. This is the converse conjectured by Nart and Novacoski. Both implications are formalized in Lean, with depth one defined by Mac Lane–Vaquié chains in a Lean 4.35 project and expressed through Okutsu sequences in a Lean 4.19 project.
Manuscript Proof and reproduction.

2.8 Discrete analysis, tilings, probability and entropy

2.8.1 The sharp Fourier bound at density one eighth

For every indicator of mean $1/8$ on a finite uniform Boolean cube, the level-one Fourier weight is at most $3/64$. A codimension-three subcube attains this bound in every dimension at least three. The proof also gives the equivalent sharp average Hamming-distance bound. It uses exact rational polynomial certificates and a dimension-independent moment identity. Fang and Huang proved this bound earlier, together with the equality cases; no originality or priority is claimed for it.
Manuscript Formal proof and reproduction.

2.8.2 A 64-point spectral tile in $\mathbb{F}_2^9$ with no periodic complement

An explicit 64-point subset E of $\mathbb{F}_2^9$ is spectral and tiles by translation, but none of its tiling complements is periodic: they are exactly the 512 translates of one aperiodic eight-point set. Hence no linear subspace complements E, and $\mathbb{F}_2^9$ fails the periodic tiling property. This answers negatively the subspace-complement part of a question of Ferguson and Iosevich; whether every spectral 64-point subset of $\mathbb{F}_2^9$ tiles remains open. The main theorem and the classification of complements are formalized in Lean 4.19.0.
Manuscript Formal proof and reproduction.

2.8.3 The small-coefficient upper half of the Keller–Klein conjecture

For a Rademacher sum $X = \sum_i v_i \varepsilon_i$ with $\sum_i v_i^2 = 1$ and every $|v_i| \le a \le 7/100$, $\Pr(|X| \le x) \le 2\Phi(x) + 2\Phi(a) - 2$ for all $x \ge 0$. This closes the range of the Keller–Klein conjecture left open by Ivanisvili's computer-assisted manuscript of 21 October 2026; together with his results it implies the full conjecture, conditionally on those unrefereed results. The theorem is formalized in Lean 4.19.0.
Manuscript Formal proof and reproduction.

2.8.4 Sharp stability and exact cap optima for Rademacher sums

Let $X = \sum_i b_i \varepsilon_i$ with $\sum_i b_i^2 = 1$ and every $|b_i| \le a$. If every signed sum has magnitude at least $x > a/2$, the core $\{i : |b_i| \ge x\}$ has odd size and the squared distance of $(|b_i|)$ to the flat vector on it is at most $2 - 2\sqrt{1-(a-x)^2}$, with equality exactly for a saturated core and one residual. At every cap, near-maximizers of the exact support-gap optimum lie within $\sqrt{2+\sqrt{2}}$ times the deficit of an admissible vector of one of the two optimal shapes, with an optimal constant, and the exact nonlinear modulus on the saturated branch is determined; these results are formalized in Lean 4.19.0. Conditionally on the completed Keller–Klein comparison, and using the Hollom–Portier tail profile, Tomaszewski's theorem and validated interval certificates, the maximal Kolmogorov distance is determined with all optimizers for $1/\sqrt{5} \le a \le 1/\sqrt{3}$.
Manuscript Proof and reproduction.

2.8.5 A quartic entropy deficit for random rankings

Let P be a probability distribution on the complete rankings of n labels. If every pairwise comparison has probability at distance at least $\varepsilon > 0$ from $1/2$, then

$$H(P) \le (1 - \varepsilon^4/144)\log(n!).$$

The formal theorem includes every finite size, including zero and one. It assumes neither independent comparisons nor transitive preferred directions. It covers both ranking lists and position-valued permutations.

Manuscript, Lean proof, and reproduction commands.

2.8.6 Sharp universal Rényi threshold for iid sums on finite Sidon sets

A sharp universal Renyi-order threshold for iid sums supported on finite strong-B2 sets. The finite-support iid theorem, zero probabilities and critical endpoint are formalized. Comparison with unverified source definitions is conditional.

Manuscript Reproduction and source context.

2.8.7 Structural selfadhesivity does not imply a numerically selfadhesive realization

An explicit five-variable conditional-independence model is structurally selfadhesive at every overlap, yet no real polymatroid inducing it is selfadhesive in Matúš's sense. This answers Question 4.18 of Boege negatively. Both parts are formalized in Lean 4.35.0-rc3 for arbitrary real polymatroids.

Manuscript Proof and reproduction.

2.9 Matroid theory

2.9.1 The asymmetric Merino–Welsh region

The theorem characterizes every nonnegative pair (x, y) for which

$$T_M(x,0)T_M(0,y) \geq T_M(1,1)^2$$

holds for all finite matroids without loops or coloops. The criterion uses two explicit scalar inequalities for every integer $k \geq 2$. The proof includes empty ground sets and finite ground sets in arbitrary ambient types. The package fetches Liu's pinned proof library from its public repository.

Manuscript Formal proof and reproduction.

2.10 Optimization

2.10.1 A narrow interval for the universal multiplier step of two-block ADMM

Two-block ADMM with relaxation only in the multiplier update converges for every constant step $0 < \tau \leq 377359/200000 = 1.886795$ on finite-dimensional convex problems whose proximal subproblems attain their minima and which have a saddle point. Fixed compact polytopes in $\mathbb{R}^{128}$, with zero objectives, admit a nonconstant periodic orbit for every $\tau \geq 3884783130026510326218902 7/205883520229331193689007 73 \approx 1.8868840$. The universal threshold therefore lies in an interval of width below 0.00009; its exact value is open. For scalar problems the universal range is exactly $0 < \tau < 2$. Lean 4.19.0 formalizes the convergence theorem and its affine-constraint versions for $0 < \tau \leq 377359/200000$, all seven finite-memory certificates and the periodic orbit; the scalar theorem is conventional.

Manuscript Proof and reproduction.

2.10.2 Escaping Nesterov flows and iterates below critical damping

For the Nesterov differential equation $X'' + (r/t)X' + \nabla F(X) = 0$ with zero initial velocity, and for the Nesterov iteration with momentum $k/(k+r)$, universal boundedness over convex objectives with a 1-Lipschitz gradient and an attained minimum holds on the Hilbert space ℓ^2 exactly when $r \geq 3$. An explicit convex planar potential has an escaping trajectory for every $0 < r \leq 2$, and its trajectories from the same start are bounded for $r > 2$. The finite-dimensional range $2 < r < 3$ remains open. The Hilbert-space and discrete thresholds and the planar escape for $1 < r \leq 2$ are formalized in Lean 4.19.0.

Manuscript Proof and reproduction.

2.10.3 Periodic obstructions to small-extrapolation Chambolle–Pock

For the unmodified Chambolle–Pock method, every step product $\kappa = \tau\sigma\|L\|^2 > 2$ admits fixed compact polytopes in $\mathbb{R}^4$ whose iteration is periodic and nonconvergent for all sufficiently small extrapolation $\theta \geq 0$. Any universal sufficient condition $\kappa < c(\theta)$ therefore has $\limsup_{\theta\downarrow 0} c(\theta) \leq 2$, below the quadratic bound $4/(1+2\theta)$. An exact universal three-cycle threshold and a rational seven-cycle are also proved. The constructions are formalized in Lean 4.19.0; the obstruction intervals in θ are very small.

Manuscript Proof and reproduction.

2.10.4 A rational five-cycle for two-block ADMM with multiplier step below two

An exact period-five orbit for multiplier-only two-block ADMM at step 199/100. Seven modules and eleven endpoint audits cover the orbit, minimizers and optimizer. Analytic regularity of the finite strongly convex extension retains a conventional proof.

Manuscript Reproduction and source context.

2.10.5 The multi-marginal converse of Minty's theorem fails in the plane

For three marginals on $\mathbb{R}^2$ with the cost $\sum_{i<j}\langle x_i, x_j\rangle$, an explicit three-point c-monotone set admits no compatible tuple with sum zero, so every maximally c-monotone extension has a sum image that omits the origin and is not convex. This answers negatively the question of Remark 2.6 of Bartz, Bauschke, Phan and Wang. The argument is formalized in Lean 4.19.0.
Manuscript Proof and reproduction.

2.10.6 The exact last-gradient bound for constant-step gradient descent in the strongly convex case

The exact final-gradient guarantee holds on every real Hilbert space for $L > 0$, $0 \leq \mu \leq L$, normalized step $0 \leq h \leq 2$ and every natural horizon. Actual one-dimensional Huber and quadratic functions attain its two branches.
Formal scope: The actual-function upper bound and normalized actual-function attainment are formalized, including every horizon and closed parameter boundary. Sharpness for arbitrary $L > 0$ and radius $R > 0$ is proved by rescaling in the manuscript, without a separate Lean wrapper.

Manuscript Proof and reproduction.

3 Theoretical computer science

3.1 Sampling algorithms and information theory

3.1.1 Critical hard-core influence bounds via nonbacktracking collisions

For every $\Delta \geq 3$, every row of the hard-core influence matrix has absolute sum $O_\Delta(\sqrt{n}\log n)$ on every n-vertex graph of maximum degree at most Δ, at every activity up to the uniqueness threshold and under every feasible pinning, with an explicit but large constant. This recovers, up to a logarithm, an $O(\sqrt{n})$ bound stated in 2025 and withdrawn in 2026. The influence bound is formalized in Lean 4.19.0; the mixing-time consequence for Glauber dynamics depends on a published proposition and is not formalized.

Manuscript Proof and reproduction.

3.1.2 Counterexamples to a best-of-n win-rate bound at every budget

For every $n \geq 2$ and $\delta > 0$, a finite positive reference distribution with largest atom at most δ violates the proposed best-of-n divergence–win-rate curve. A four-response best-of-two example gives exact probabilities and a strict logarithmic certificate. The all-budget proof uses finite interval bounds and gives a quantitative positive gap. It does not determine the sharp replacement curve.

Scope: Disproof for every budget n≥2 and every positive atom cap.

Manuscript Formal statement and prior work.

3.2 Boolean functions and decision trees

3.2.1 Logarithmic influence bounds and variance counterexamples for Boolean decision-tree covariance

Let T be a finite Boolean decision tree under uniform independent bits. No coordinate repeats on a root-to-leaf path. Let $k_i(T)$ count all query nodes labelled i, and let $I_i(f)$ be the probability that flipping coordinate i changes the output. The weighted sum of absolute child covariances satisfies

$$\sum_v 2^{-d(v)} \, |\mathrm{Cov}(f_{v,+}, f_{v,-})| \leq \sum_i I_i(f) \log_2 \max\{1, k_i(T)\}.$$

Here $d(v)$ is the node depth, and the two child functions use the same input cube. The read number counts nodes in the entire tree, not queries along one path.

For a read-k tree, the bound is $I(f) \log_2 k$. The dummy-prefix construction of Wan, Wright, and Wu attains equality when k is a power of two. The logarithmic potential is due to Shalev. A four-variable counterexample has covariance 5/4 and variance 3/4. A nine-variable counterexample has covariance 4863/4096 and variance 3007/4096, with every query relevant. Both refute the coefficient-one logarithmic variance proposal for a given tree.

Scope: Covariance inequalities, an equality family, and exact finite witnesses. No Fourier-entropy theorem is formalized.

Manuscript Prior work Formal proof and reproduction.

3.3 Fair division

3.3.1 Ordered-instance allocation transfer with exact extra-copy preservation

A set-valued ordered-instance transfer preserving utility improvement, the exact extra-copy count, per-good caps and coverage. The transfer theorem is formalized. Existence of a fair allocation before transfer is outside the result; external source comparison is provisional.

Manuscript Reproduction and source context.

3.4 Proof complexity

3.4.1 Unbounded coalesced-mass certificates for resolution trees

A coalesced dyadic clause-mass functional has exact supremum n+1 on the stated unrestricted resolution-tree class and is unbounded across dimensions. The internally defined resolution-tree functional and proof class are formalized. No unconditional identification with an external conjecture or a regular proof class is claimed.

Manuscript Reproduction and source context.

4 Mathematical physics

4.1 Stochastic thermodynamics

4.1.1 A three-state thermal counterexample to a heat-current kurtosis conjecture

A three-state rate model with two positive-temperature baths satisfies local detailed balance. Every total off-diagonal rate is positive. The tilted generator records heat transferred to the cold bath. Its dominant eigenvalue branch is four times continuously differentiable near zero and has a larger real part than every distinct complex eigenvalue. The exact derivative ratio is

$$\frac{C_4}{C_2} = -\frac{3470947982622836}{176627429595846691} < 0,$$

where C_n is the nth derivative at zero. This refutes Eq. (19) of Ptaszyński, *Physical Review E* **106**, 024119 (2022), in the paper's finite thermal spectral formulation. The third derivative is positive, so the witness does not refute the separate skewness conjecture.

Scope: Finite rate matrices, local detailed balance, complex eigenvalues, and derivatives of a local dominant branch. The stochastic long-time theorem that identifies the branch with a cumulant generating function is not formalized. No minimal-state or empirical validation claim is made.

Manuscript Prior work Formal proof and reproduction.

4.1.2 A sharp universal skewness bound for unicyclic currents

The net marked-edge current on every finite bidirectional Markov ring with at least three states, positive rates and unequal forward and reverse rate products satisfies $-1/16 \le \mathrm{C}_3/\mathrm{C}_1 \le 1$. The two-channel two-state case has the stronger interval [1/4,1]. This ratio is not standardized skewness.

Formal scope: The interval for actual tilted-matrix derivatives and a local branch is formalized. Identification with long-time stochastic cumulants and sharpness have conventional proofs. Lean does not construct a continuous-time sample-path probability space or a separate optimizing-sequence theorem.

Manuscript Proof and reproduction.

4.2 Quantum information

4.2.1 Universal entanglement embezzlement

A target-independent catalyst of local dimension 2^{4L} gives root infidelity at most $125m/L^2$ for every Schmidt probability vector on m entries and $L \ge 1$. Zero entries are permitted. The proof constructs local basis permutations and gives their finite complex Hilbert-space interpretation. It does not separately formalize arbitrary changes to an external Schmidt basis, gate efficiency, or catalyst reuse.

Manuscript Formal proof and reproduction.

4.2.2 The four-qubit Higuchi–Sudbery entropy bound

For every normalized pure state of four qubits, the average von Neumann entropy of the three balanced two-qubit cuts is at most $\log 2 + \frac{1}{2}\log 3$. The Higuchi–Sudbery state attains this value. The formal proof uses all sixteen complex amplitudes, the actual reduced matrices and their eigenvalues, an exact sum-of-squares identity, and an explicit attaining state. It does not classify all maximizers or cover mixed states.

Manuscript, Lean proof, and reproduction commands.

4.2.3 A pure-state matched-basis information inequality violation in dimension five

A dimension-five pure-state matched-basis information inequality violation, with a conventional all-prime companion. The dimension-five identical-local-basis theorem is formalized. The prime-family extension is conventional; arbitrary or independently optimized basis choices are outside the result.

Manuscript Reproduction and source context.

4.2.4 Exact three-setting correlation bounds at fixed steering centre

The exact least upper bound of the optimized three-setting squared correlation value is determined for every two-qubit steering-centre length. The threshold for a value greater than one is $c < 2/3$. Attainment, nonattainment and the singular marginal boundary are included.

Formal scope: Actual two-qubit density matrices, canonical filters, Pauli correlations, the exact least upper bound, attainment and nonattainment, and fully entangled fraction are formalized. The steering-ellipsoid and optimized CJWR measurement interpretations use cited external correspondences; general steerability is outside the result.

Manuscript Proof and reproduction.

4.2.5 Equality, rigidity and full-ellipsoid CHSH envelopes at fixed steering-ellipsoid centre

For two-qubit states with steering-ellipsoid centre length c and Bob Bloch length r, the CHSH bound $\beta^2/4 \leq 1 + (1 - r^2)(1 - 2c)$ and its attaining value are prior work: the Omega Institute's public repository recorded a claimed Lean proof of Milne et al.'s Conjecture 1 containing this bound on 14 October 2026, and Zhang et al. (2020) computed the attaining value; no priority is claimed for them. For $0 < c < 1$ the extremizers form one local-unitary class, a filtered maximally obese state with $|b|^2 = b_z$. Near-extremal states are close in trace norm to exact extremizers, with a uniform exponent $1/4$ that is optimal. Exact fixed-radius CHSH envelopes are given for every axial and every centred steering ellipsoid, with an exact CHSH-breaking criterion for axial qubit channels.

Formal scope: Lean 4.19.0 proves, for every admissible pair (c, r), that an explicit complex density matrix with centre $(0, 0, c)$ and Bob length r has CHSH supremum equal to $2\sqrt{1 + (1 - 2c)(1 - r^2)}$, and checks two further witnesses and the scalar axial spectral optimization; the equality classification, rigidity and general filtering steps are conventional proofs.

Manuscript Proof and reproduction.

4.2.6 Optimal entanglement witnesses under ellipsoid containment: the Milne–Jennings–Rudolph conjecture and its chirality correction

Milne, Jennings and Rudolph conjectured that if $E^\star$ is an ellipsoid in the Bloch ball and S is the set of two-qubit entanglement witnesses whose ellipsoid lies inside $E^\star$, then the optimal witnesses in S are represented by $E^\star$. The conjecture holds as stated: every optimal member of S has ellipsoid $E^\star$. The stronger reading, that every witness with ellipsoid $E^\star$ is optimal, fails: for $0 < p < 1/2$ the witnesses W_p and W_{1-p} of their Example 2 share a full-dimensional Class D ellipsoid and W_{1-p} is strictly finer. For a full-dimensional target the optimal witnesses are exactly the right-handed representatives, and for a planar target all representatives. Lean proves the counterexample for genuine complex matrices and every $0 < p < 1/2$, with further matrix and geometric ingredients; the classification for arbitrary targets is a conventional proof using the published decomposability and Lorentz normal-form theorems.

Manuscript Proof and reproduction.

4.2.7 The full correlation range, filtering frontier and volume trade-off at fixed steering centre

For two-qubit states with steering-ellipsoid centre length c and Bob Bloch length $r < 1$, the values of $F_3^2 = \|T\|_F^2$ form exactly the interval $[\max(0, c(1+r)-1)^2,\ (1-r^2)(3-4c)+(c+(1-c)r)^2]$, with separable minimizers. For canonical states with radial support half-width t and squared correlation τ, the largest steering-ellipsoid volume is $(4\pi/3)(t/2)\min(\tau - t^2,\ 1+2t+2t^2-c^2-\tau)$, attained at every feasible triple. A filter succeeding with probability at least $p \in [1/2, 1]$ can reveal $F_3 > 1$ from a canonical input exactly when $c < 2/(1+p+\sqrt{(1-p)^2+2})$. Every one-sided filtering orbit is solved and the maximizers of the fixed-centre envelope are classified.
Formal scope: Lean 4.19.0 proves the joint interval, the probability frontier from actual Kraus contractions, the orbit solution, the maximizer classification with trace-norm rigidity and the volume maxima in a fixed marked-axis convention, for actual complex density matrices; a whole-project audit covers all 70 modules. The flat-branch exponents, the volume-maximizer orbit classification and the CJWR measurement interpretation are conventional. The Omega Institute's public repository records an earlier claimed Lean proof of the fully entangled fraction bound included unchanged from the previous package; no priority is claimed for it.

Manuscript Proof and reproduction.

4.2.8 Exact one-sided filtering of the two-qubit F_3 functional

Ducuara, Susa and Skrzypczyk defined one-sided hidden correlation measures, in which only one party applies an invertible local filter, and noted that no closed formulas for them were known. For both one-sided F_3 measures of their Equation (10), and every two-qubit state, the squared value is $\beta + \eta + c^T Q(\eta I - Q)^\dagger c$ in the other party's steering-ellipsoid centre c and shape Q, where η is the larger of $\operatorname{tr} Q - \|c\|^2$ and the largest real eigenvalue of an explicit 6×6 matrix. Every optimal filter and attainment by invertible filters are described, a value above one has a rational formula, and a strictly positive rational state has an optimal value with Galois group S_6, so no formula in radicals exists. The one-sided CHSH measures of their Equation (9) are not treated.
Formal scope: Lean proves the main theorem for actual complex 4×4 density matrices and invertible complex filters on either side, with no rank assumptions, and audits every project declaration. The rational Bloch-coordinate expressions, the Galois obstruction and the operational extensions are conventional proofs with exact and numerical checks.

Manuscript Proof and reproduction.

4.2.9 The one-sided CHSH measure under local filtering: exact supremum, violation criterion and a least-eigenvalue formula

Ducuara, Susa and Skrzypczyk defined the one-sided hidden CHSH measures, the largest normalized CHSH value when only one party applies an invertible local filter, and noted that no closed formula was known. For every two-qubit state, after whitening the filtering party, the supremum is the maximum of an explicit quadratic over a compact set, in general not attained. With whitened data c, D and $A = DD^{\mathsf{T}}$, $K = A - \|c\|^2 I$, $C = [Dc]_\times$, $\alpha = \operatorname{tr} A - \|c\|^2$, one-sided violation requires $\alpha > 1$, and there the squared supremum is $\operatorname{tr} A - \lambda_{\min}(A + CK^{-1}C)$, attained by an explicit invertible filter; violation occurs exactly when this exceeds 1. States with $\alpha \le 1$ have value at most 1, given by at most four algebraic candidates. The quasi-distillable family $p\,\psi^- + (1-p)|00\rangle\langle 00|$ has one-sided value $\max\{1, 2p/(2-p)\}$.
Formal scope: Lean 4.19.0 proves, for every positive semidefinite trace-one complex 4×4 matrix, every invertible complex filter on either party and arbitrary traceless Hermitian involutive qubit observables, the exact suprema, the least-eigenvalue formula, the violation criterion, invertible attainment and the finite-candidate theorem; a whole-project audit covers all 32 modules. The

secular-root boundary radius, the polynomial activation tests, the rank-two formula and the success-probability trade-offs are conventional. The Omega Institute's public repository records an earlier claimed Lean proof of Milne's Conjecture 1 that uses the same plane-purity identity and filtering covariance and, with Result 1 of Ducuara et al., already gives the inaccessible interval $0 < p \leq 2/3$; no priority is claimed for these.

Manuscript Proof and reproduction.

4.2.10 Saturating the steering-centre bound on the fully entangled fraction

The bound $f \leq 1 - c/2$ on the fully entangled fraction of a two-qubit state with steering-ellipsoid centre length c is Conjecture 2 of Milne, Jennings, Jevtic and Rudolph; it is prior work, recorded with a claimed Lean proof by the Omega Institute on 8 October 2026 and proved in the steering-bounds package, and no priority is claimed for it. With Bob's Bloch length $b < 1$ also fixed, the sharp bound is $f \leq \left[\sqrt{1+b} + \sqrt{(1-c)(1-b)}\right]^2/4$, attained for every pair and equal to $1 - c/2$ exactly when $b = c/(2-c)$. Each frontier value is attained by one local-unitary orbit; for each c the states with $f = 1 - c/2$ form one orbit, of rank two for $0 < c < 1$; and the best constant in $((2-c)b - c)^2 \leq K(1 - c/2 - f)$ is $4(1 + \sqrt{1-c})^2$. In every dimension a prescribed-Schmidt overlap bound in the two marginal spectra is sharp and its saturators have a complete normal form, with optimizer-rank bounds, ellipsoid-shape refinements, the tensor law $\mathrm{SN}(\rho \otimes \sigma) = r\,\mathrm{SN}(\sigma)$, a qutrit state with equal G-concurrence roofs whose tensor square has G-concurrence zero, negativity optimization and the exact collective coupling capacity.
Formal scope: Lean 4.32.0 proves the joint frontier, its sharpness for every pair, the stability bound and the optimality of its constant, the general prescribed-Schmidt bound, the fixed-marginal reduction and the rank ceilings for every optimizer, for actual complex matrices, and replays the Omega proof of the known bound with attribution; a whole-project audit covers every declaration. The equality classifications, the saturation normal form, the tensor laws and the negativity and capacity theorems are conventional proofs with exact certificates and numerical checks.

Manuscript Proof and reproduction.

4.2.11 The three-state root-fidelity entropy bound

The entropy of the weighted three-by-three root-fidelity matrix bounds the Holevo quantity of every three-state ensemble in arbitrary finite dimension. The result includes mixed and singular states and zero probabilities.
Formal scope: The unrestricted three-state ensemble theorem is formalized for actual positive semidefinite complex matrices of trace one, nonnegative weights summing to one, spectral entropy and nuclear-norm root fidelity. Credited QICLean foundations and prior-work adapters retain their attribution.

Manuscript Proof and reproduction.

4.2.12 Exact conditional contraction for Pauli channels and depolarization in every dimension

The conditional hockey-stick contraction coefficient is determined for every stochastic Pauli qubit channel and for depolarization throughout its physical parameter interval in every dimension $d \geq 2$. The formulas include all boundary cases and every parameter $\gamma \geq 1$, with arbitrary finite quantum references.
Formal scope: Actual channel constructors, unrestricted upper bounds and matching density-state supremum witnesses are formalized. Unital-qubit normal-form existence is cited. Erasure-simulation and ensemble-guessing consequences retain conventional proofs.

Manuscript Proof and reproduction.

4.2.13 Strong minimum-rank multiplicativity from unextendible product constraints

A complex matrix subspace annihilated by an unextendible family of product functionals doubles minimum rank under tensor products with any partner. For the Tiles bound entangled state ρ, the Schmidt number of $\rho^{\otimes N}$ is exactly 2^N for every $N \geq 1$, extending the known two-copy case. The general theorem and the all-copy formula are formalized in Lean 4.19.0.

Manuscript Proof and reproduction.

4.2.14 No universal finite collapse of the upper PPT entanglement-cost hierarchy

For every $q \geq 1$ there is a full-rank Bell-diagonal state in local dimension $2^{(2q+4)4^q}$ with $\kappa_{q+1}(\rho) < \kappa_q(\rho)$ in the upper hierarchy of Lami, Mele and Regula, so no fixed finite upper level is exact for all states. Brinster et al. had proved the corresponding statement for the lower hierarchy. The strict separation for the actual semidefinite programs is formalized in Lean 4.19.0.

Manuscript Proof and reproduction.

5 Manuscripts and proof packages

The repository contains the manuscripts, source files and supporting proof packages. The README highlights eleven standout results, in order of significance: the small-coefficient Keller–Klein bound, the strong-convexity gradient bound, the ADMM multiplier-step interval, the upper PPT hierarchy, depth-one Artin–Schreier towers, matrix-pencil rank distance, nonnegative rank of Kronecker squares, finite real q-permanent counterexamples, the density-one-eighth Fourier bound, subcritical Nesterov flows and iterates, and universal embezzlement. Results with earlier or overlapping work recorded in the priority and novelty record are not selected, except the density-one-eighth bound, which was committed publicly before the first public appearance of an independent proof. Their manuscripts state the autonomous AI research disclosure and exact verification scope. The catalogue links all retained results and reproduction instructions.

6 Verification

Lean checks the proofs against their formal statements. In the shared Lean 4.35.0 library, separate Challenge modules state the intended claims. Comparator checks these against their proved Solutions. The verification record identifies the completed runs of Lean, NanoDa, and con-ron for those seven families. The standalone study packages have their own pinned Lean environments and stated proof boundaries. Source-integrity checks are distinguished from completed compilation records. Their audit logs state the permitted axioms. Exact arithmetic programs check the accompanying finite certificates. The shared Comparator results do not cover those packages. Independent exact and numerical programs test finite certificates and examples. These finite computations do not replace the general proofs.